
❏ Operating System: Functions – MCQ Practice Set (30 Questions)

1. What is the main function of the operating system?

- A. Editing text
- B. Allocating system resources
- C. Translating code
- D. Browsing the internet

❏ Answer: B

Explanation: OS manages hardware resources like CPU, memory, I/O.

2. Which part of the OS interacts directly with the hardware?

- A. Shell
- B. GUI
- C. Kernel
- D. Compiler

❏ Answer: C

Explanation: Kernel handles low-level hardware operations.

3. What does the OS use to manage multiple tasks at the same time?

- A. Virtual memory
- B. Scheduling
- C. Threading
- D. Paging

❏ Answer: B

Explanation: Scheduler decides which process runs when.

4. The program responsible for starting the OS when the computer is turned on is:

- A. Shell
- B. Loader
- C. BIOS
- D. Bootloader

☐ **Answer: D**

Explanation: Bootloader loads the OS kernel into memory.

5. Which of the following is not a function of an OS?

- A. Memory Management
- B. Process Management
- C. Device Management
- D. Data Mining

☐ **Answer: D**

Explanation: Data mining is an application-level function.

6. The OS component responsible for handling I/O devices is:

- A. File Manager
- B. Device Manager
- C. Scheduler
- D. Loader

☐ **Answer: B**

7. Which function ensures no two processes access the same data at the same time?

- A. Security
- B. Synchronization
- C. Protection
- D. Scheduling

☐ **Answer: B**

8. Memory management deals with:

- A. Compiling programs
- B. Allocating memory blocks to programs
- C. Controlling devices
- D. Scheduling tasks

☐ **Answer: B**

9. Process management includes all EXCEPT:

- A. Creating processes
- B. Terminating processes
- C. Translating source code
- D. Scheduling processes

☐ **Answer: C**

10. Which OS function involves tracking file location, status, and permissions?

- A. Memory Management
- B. File Management
- C. Process Management
- D. Security

☐ **Answer: B**

11. Deadlock prevention is part of:

- A. Memory Management
- B. File System
- C. Process Management
- D. Device Management

☐ **Answer: C**

12. The OS ensures data security via:

- A. Authentication
- B. Compression
- C. Sorting
- D. Scheduling

☐ **Answer: A**

13. In multitasking, the OS switches between:

- A. Hard drives
- B. Users
- C. Programs

D. Processes

☐ **Answer: D**

14. Virtual memory is managed by:

A. Kernel

B. Application software

C. File system

D. BIOS

☐ **Answer: A**

15. Which of the following schedules process execution?

A. File Manager

B. Compiler

C. Scheduler

D. Buffer

☐ **Answer: C**

16. The OS maintains time-sharing using:

A. Interrupts

B. Paging

C. Round-robin scheduling

D. Virtualization

☐ **Answer: C**

17. Time taken for OS to switch between processes is:

A. Access time

B. Turnaround time

C. Context switch time

D. Seek time

☐ **Answer: C**

18. OS provides user interface through:

- A. APIs
- B. GUI or CLI
- C. HTTP
- D. DNS

☐ **Answer: B**

19. Which OS function improves CPU utilization?

- A. Memory management
- B. Scheduling
- C. Device control
- D. Security

☐ **Answer: B**

20. Paging is related to:

- A. Process synchronization
- B. Memory management
- C. I/O scheduling
- D. File handling

☐ **Answer: B**

21. What manages software interrupts?

- A. Loader
- B. Scheduler
- C. Kernel
- D. Compiler

☐ **Answer: C**

22. The OS maintains logs and history for:

- A. Accounting
- B. Paging
- C. Scheduling

D. Booting

☐ **Answer: A**

23. File permissions are part of:

A. Memory management

B. File management

C. Protection

D. Process management

☐ **Answer: C**

24. CPU idle time is minimized using:

A. Round robin

B. Virtual memory

C. Preemptive multitasking

D. Disk defragmentation

☐ **Answer: C**

25. Which OS manages real-time tasks?

A. Unix

B. Real-Time OS

C. DOS

D. Android

☐ **Answer: B**

26. Which is not a type of OS?

A. Batch

B. Time-sharing

C. Distributed

D. Sequential

☐ **Answer: D**

27. Which allows communication between user and OS?

- A. Driver
- B. Kernel
- C. Shell
- D. BIOS

☐ **Answer: C**

28. Swapping is related to:

- A. Scheduling
- B. Device Management
- C. Memory Management
- D. File Management

☐ **Answer: C**

29. OS loads and executes programs through:

- A. BIOS
- B. Interpreter
- C. Loader
- D. Assembler

☐ **Answer: C**

30. Which of these OS functions allows resource sharing among users?

- A. Synchronization
- B. Multiprogramming
- C. Swapping
- D. Deadlock detection

☐ **Answer: B**

Here are **MCQs with options** and **previous years' questions** for the topic:

Operating System – Main Functions of Operating Systems

☐ **MCQs (with options)**

1. Which of the following is not a function of an operating system?

- A. Memory Management
- B. Compiler Design
- C. File Management
- D. Process Management

Answer: B. Compiler Design

2. The part of the operating system that manages hardware and low-level functions is called:

- A. Shell
- B. Kernel
- C. Command Interpreter
- D. Device Driver

Answer: B. Kernel

3. Which operating system function is responsible for controlling how programs are executed?

- A. File Management
- B. Device Management
- C. Process Management
- D. Memory Allocation

Answer: C. Process Management

4. What function of the OS helps multiple programs run at the same time?

- A. Booting
- B. Multitasking
- C. Debugging
- D. Compiling

Answer: B. Multitasking

5. Operating system manages which of the following resources?

- A. CPU
- B. Memory
- C. I/O Devices

D. All of the above

Answer: D. All of the above

6. Which of the following is not handled by an operating system?

A. Security

B. Job accounting

C. Internet browsing

D. Deadlock handling

Answer: C. Internet browsing

▢ **Previous Years' CUET PG Questions (from Operating Systems – Functions)**

Q1. (CUET PG 2022)

Which of the following is NOT a major function of the Operating System?

A. Managing hardware

B. Providing user interface

C. Compiling code

D. File system management

Answer: C. Compiling code

Q2. (CUET PG 2023)

The main job of the operating system is to:

A. Provide utilities to the user

B. Manage computer resources

C. Allow user programs to interact with each other

D. Manage web services

Answer: B. Manage computer resources

Q3. (CUET PG 2021)

Which of the following functions are performed by the OS?

(i) Process scheduling

(ii) Memory allocation

(iii) Email filtering

- (iv) File handling
- A. (i), (ii), and (iv)
 - B. (i), (iii), and (iv)
 - C. (ii) and (iv) only
 - D. All of the above

Answer: A. (i), (ii), and (iv)

Great! Here's a **comprehensive MCQ practice set (30 questions)** on the **Processes** topic from Operating Systems, including previous CUET PG-style questions.

☐ Operating System - Topic: Processes

1. A process is:

- A. A program in execution
- B. A thread
- C. A compiled program
- D. A file in memory

☐ **Answer:** A

Explanation: A process is a running instance of a program.

2. Which of the following is not a process state?

- A. New
- B. Running
- C. Blocked
- D. Terminated
- E. Executable

☐ **Answer:** E

3. What is a PCB?

- A. Process Control Block
- B. Program Control Base
- C. Process Code Binary
- D. Program Configuration Block

☐ **Answer:** A

Explanation: PCB stores process-related information like PID, state, registers, etc.

4. Which OS component schedules processes?

- A. Dispatcher
- B. Loader
- C. Shell
- D. Compiler

☐ **Answer:** A

5. Which of the following allows multiple processes to share CPU time?

- A. Multithreading
- B. Swapping
- C. Context Switching
- D. Interrupts

☐ **Answer:** C

6. A process is said to be in a blocked state when:

- A. It is ready for execution
- B. It is waiting for CPU
- C. It is waiting for an event like I/O
- D. It is terminated

☐ **Answer:** C

7. What causes a context switch?

- A. I/O request
- B. Time slice expiration
- C. Interrupt

D. All of the above

☐ **Answer: D**

8. Which is not stored in a PCB?

A. Program Counter

B. Process State

C. Memory Map

D. File System

☐ **Answer: D**

9. The parent process creates a child process using:

A. exec()

B. fork()

C. start()

D. begin()

☐ **Answer: B**

Explanation: fork() is used in Unix-based systems to create child processes.

10. When a process finishes executing, it moves to:

A. Ready

B. Running

C. Terminated

D. Blocked

☐ **Answer: C**

11. Which system call replaces the current process image with a new one?

A. fork()

B. wait()

C. exec()

D. spawn()

☐ **Answer: C**

12. What is the parent process of all processes in Unix?

- A. init
- B. bash
- C. sh
- D. kernel

☐ **Answer: A**

13. The system call used to create a new process is:

- A. fork()
- B. create()
- C. new()
- D. exec()

☐ **Answer: A**

14. A child process can:

- A. Be created by any process
- B. Access its parent's code
- C. Run independently
- D. Be created only by init

☐ **Answer: C**

15. Zombie process is:

- A. A process in waiting
- B. A terminated process that has not been cleaned
- C. A suspended process
- D. A deadlock

☐ **Answer: B**

16. Orphan process is:

- A. Process without a PCB
- B. Child process with no parent alive
- C. Process in ready state

D. None of the above

☐ **Answer: B**

17. The command to list all processes in Unix/Linux is:

A. ps

B. kill

C. showproc

D. tasklist

☐ **Answer: A**

18. A process can be in which of these states?

A. New, Ready, Running

B. Blocked, Suspended, Terminated

C. All of the above

☐ **Answer: C**

19. The time taken to switch from one process to another is:

A. Execution time

B. Switching time

C. Context switching time

D. CPU burst time

☐ **Answer: C**

20. Process scheduling improves:

A. RAM usage

B. CPU utilization

C. Disk access

D. Power consumption

☐ **Answer: B**

21. Which of these is handled during context switch?

- A. Stack pointer
- B. Program counter
- C. Registers
- D. All of the above

☐ **Answer: D**

22. Thread is:

- A. Independent process
- B. Lightweight process
- C. System call
- D. Kernel module

☐ **Answer: B**

23. Which statement is true for processes and threads?

- A. Threads share resources of the process
- B. Processes share threads
- C. Threads are heavier than processes
- D. Threads have different memory

☐ **Answer: A**

24. Long-term scheduler:

- A. Selects from ready queue
- B. Controls degree of multiprogramming
- C. Manages I/O
- D. Schedules disk

☐ **Answer: B**

25. The process that controls all others is:

- A. User process
- B. init process
- C. kernel

D. shell

☐ Answer: B

26. Which process gets CPU time first in FCFS?

A. Shortest

B. Longest

C. First arrived

D. Highest priority

☐ Answer: C

27. Process ID (PID) is:

A. Memory address

B. Kernel function

C. Unique number given by OS

D. Input from user

☐ Answer: C

28. Process synchronization is needed to:

A. Manage memory

B. Control process execution order

C. Avoid data inconsistency

D. Improve battery life

☐ Answer: C

29. System call to terminate a process is:

A. stop()

B. exit()

C. destroy()

D. kill()

☐ Answer: B

30. A cooperating process is:

- A. One that shares data with others
- B. One that is always terminated
- C. One that runs solo
- D. One that avoids sharing

☐ **Answer: A**

☐ **CUET PG Previous Year Process Questions**

Q1. (CUET PG 2022)

Which of the following is true about context switching?

- A. Only occurs during file transfer
- B. Happens when CPU moves from one process to another
- C. Occurs only in batch processing
- D. Never used in time-sharing

☐ **Answer: B**

Q2. (CUET PG 2023)

Which of these is not a process state?

- A. Waiting
- B. Ready
- C. Completed
- D. Blocked

☐ **Answer: C**

Q3. (CUET PG 2021)

What is the role of a Process Control Block (PCB)?

- A. Handle user interface
- B. Maintain memory addresses
- C. Store process information
- D. Schedule CPU

☐ **Answer: C**

Here is a **comprehensive MCQ practice set (30 questions)** on the topic **Threads** from the Operating Systems syllabus for CUET PG MCA, including **CUET previous year-style questions** with explanations.

□ Operating System – Topic: Threads

1. A thread is also called:

- A. Mini process
- B. Sub-process
- C. Lightweight process
- D. Micro kernel

□ Answer: C

Explanation: A thread is a lightweight process with its own execution path but shared memory.

2. Which of the following is shared among threads in the same process?

- A. Register values
- B. Stack
- C. Program counter
- D. Address space

□ Answer: D

3. Threads of the same process do not share:

- A. Code section
- B. Data section
- C. Heap
- D. Stack

□ Answer: D

Explanation: Each thread has its own stack.

4. Multithreading increases:

- A. CPU idle time
- B. Memory usage

- C. Responsiveness and throughput
- D. Disk fragmentation

☐ **Answer: C**

5. Which of the following does a thread have?

- A. Its own address space
- B. Its own stack and program counter
- C. Its own data segment
- D. Its own heap

☐ **Answer: B**

6. In a multithreaded process, all threads share:

- A. Instruction pointer
- B. Stack
- C. CPU registers
- D. Global variables

☐ **Answer: D**

7. The main advantage of threads over processes is:

- A. Less overhead
- B. Isolation
- C. Increased memory use
- D. Kernel involvement

☐ **Answer: A**

8. User-level threads are:

- A. Managed by kernel
- B. Faster than kernel-level threads
- C. Scheduled by hardware
- D. Created only by OS

☐ **Answer: B**

9. Kernel-level threads are:

- A. Invisible to OS
- B. Scheduled by OS kernel
- C. Managed by user programs
- D. Always faster than user threads

☐ **Answer: B**

10. Thread switching within the same process is:

- A. More expensive than process switching
- B. Done by the loader
- C. Faster than process switching
- D. Done by BIOS

☐ **Answer: C**

11. Multithreading is useful when:

- A. Tasks are sequential
- B. Tasks are independent and can run concurrently
- C. There's only one task
- D. CPU is idle

☐ **Answer: B**

12. Threads are primarily used for:

- A. Reducing power
- B. Parallel execution
- C. Disk optimization
- D. File compression

☐ **Answer: B**

13. POSIX threads are also known as:

- A. System threads
- B. Kernel threads
- C. Pthreads

D. Multithreads

☐ **Answer: C**

14. Which of the following does not use threads?

- A. Web server handling multiple clients
- B. Word processor spell-checker and editing
- C. Single-core BIOS boot
- D. Video player buffering and decoding

☐ **Answer: C**

15. What causes race conditions in threads?

- A. Slow hardware
- B. Shared resources without proper synchronization
- C. Lack of RAM
- D. No GUI

☐ **Answer: B**

16. Which language supports built-in threading?

- A. HTML
- B. Python
- C. SQL
- D. CSS

☐ **Answer: B**

17. Which system call creates a thread in POSIX?

- A. fork()
- B. create()
- C. pthread_create()
- D. exec()

☐ **Answer: C**

18. Daemon threads are:

- A. Used by BIOS
- B. Background threads that die with the process
- C. Always terminated manually
- D. Used in security only

☐ **Answer: B**

19. Thread pool is:

- A. A process with many threads
- B. A collection of pre-created threads ready to execute tasks
- C. A type of memory
- D. A user process

☐ **Answer: B**

20. Thread priority affects:

- A. Memory allocation
- B. Execution order
- C. Code compilation
- D. Stack size

☐ **Answer: B**

21. Threads improve performance by:

- A. Pausing I/O
- B. Wasting CPU time
- C. Doing multiple tasks in parallel
- D. Sleeping often

☐ **Answer: C**

22. If one thread blocks on I/O, the others:

- A. Are terminated
- B. Continue execution
- C. Also block

D. Are suspended

☐ **Answer: B**

23. Thread lifecycle does not include:

A. New

B. Ready

C. Executed

D. Terminated

☐ **Answer: C**

24. Thread-safe code:

A. Uses global variables

B. Avoids synchronization

C. Works correctly with multiple threads

D. Crashes with multithreading

☐ **Answer: C**

25. Which threading model maps many user threads to one kernel thread?

A. One-to-one

B. Many-to-many

C. Many-to-one

D. Two-to-one

☐ **Answer: C**

26. Java threads are managed using:

A. exec()

B. Thread class and Runnable interface

C. BIOS

D. JavaCompiler

☐ **Answer: B**

27. One advantage of multithreaded programming is:

- A. Increased complexity
- B. Isolation of tasks
- C. Lower performance
- D. Simplified resource sharing

☐ **Answer: D**

28. Which operation is not thread-safe by default?

- A. StringBuffer in Java
- B. printf in C
- C. Java's Hashtable
- D. ArrayList in Java

☐ **Answer: D**

29. A race condition can be avoided using:

- A. Busy waiting
- B. Critical section and mutex
- C. Delays
- D. Paging

☐ **Answer: B**

30. Which of the following is a true statement?

- A. Threads run slower than processes
- B. Threads use more memory than processes
- C. Threads are isolated from each other
- D. Threads share process resources

☐ **Answer: D**

☐ **CUET PG Previous Year Thread Questions**

Q1. (CUET PG 2023)

Which of the following is shared between threads in the same process?

- A. Stack

- B. Program Counter
- C. Heap
- D. Registers

☐ **Answer: C**

Q2. (CUET PG 2022)

A thread has its own:

- A. Code
- B. Stack
- C. Data segment
- D. Address space

☐ **Answer: B**

Q3. (CUET PG 2021)

Which of the following is true regarding multithreading?

- A. Each thread has its own address space
- B. Threads do not share global data
- C. Threads are lighter than processes
- D. Threads cannot run concurrently

☐ **Answer: C**

Great! Here's a full **MCQ practice set (30 questions)** on **Interprocess Communication (IPC)** for CUET PG MCA, along with previous year-style questions and brief explanations.

☐ **Operating System – Topic: Interprocess Communication (IPC)**

1. Interprocess Communication is used for:

- A. Memory allocation
- B. Communication between processes
- C. Compiling code
- D. Creating new files

☐ **Answer: B**

2. IPC allows processes to:

- A. Share code
- B. Communicate and synchronize
- C. Increase memory
- D. Switch users

☐ **Answer: B**

3. Which of the following is NOT a method of IPC?

- A. Pipes
- B. Message Passing
- C. Signals
- D. Memory Allocation

☐ **Answer: D**

4. Which IPC mechanism allows unidirectional communication?

- A. Sockets
- B. Shared memory
- C. Named pipes
- D. Anonymous pipes

☐ **Answer: D**

5. A pipe provides:

- A. Bidirectional data transfer
- B. Unidirectional data flow between processes
- C. Memory mapping
- D. CPU allocation

☐ **Answer: B**

6. Which of the following supports bidirectional communication?

- A. Shared Memory
- B. Anonymous Pipes

- C. Message Queues
- D. Named Pipes (FIFOs)

☐ **Answer: D**

7. The fastest IPC mechanism is:

- A. Pipes
- B. Message queues
- C. Shared memory
- D. Sockets

☐ **Answer: C**

8. What is a potential risk in shared memory IPC?

- A. Race condition
- B. Port blocking
- C. Context switching
- D. Paging

☐ **Answer: A**

9. Shared memory requires:

- A. Mutex or semaphores for synchronization
- B. Context switching
- C. CPU time slice
- D. Memory mapping

☐ **Answer: A**

10. Message queues allow processes to:

- A. Share memory addresses
- B. Communicate using fixed-size messages
- C. Replace system calls
- D. Copy code segments

☐ **Answer: B**

11. Which of the following is not a synchronization tool?

- A. Semaphore
- B. Mutex
- C. Pipe
- D. Monitor

☐ **Answer: C**

12. Signals in IPC are used to:

- A. Run instructions
- B. Notify processes of events
- C. Transfer memory
- D. Format drives

☐ **Answer: B**

13. Which function is used to create a pipe in UNIX?

- A. open()
- B. pipe()
- C. fork()
- D. exec()

☐ **Answer: B**

14. Message passing is useful when:

- A. Processes share memory
- B. Processes are on different systems
- C. Synchronization is not needed
- D. Same thread communicates

☐ **Answer: B**

15. Sockets are used in:

- A. Local IPC only
- B. Remote IPC (networked systems)
- C. Internal interrupts

D. Compilers

☐ Answer: B

16. The system call used to send a signal in UNIX is:

A. fork()

B. kill()

C. wait()

D. sleep()

☐ Answer: B

17. Semaphores are primarily used for:

A. Process creation

B. Process termination

C. Synchronization

D. Scheduling

☐ Answer: C

18. A critical section problem can occur when:

A. Processes share resources

B. Processes run in isolation

C. OS loads the bootloader

D. Compiler compiles

☐ Answer: A

19. Peterson's algorithm is used to solve:

A. CPU scheduling

B. Deadlock detection

C. Synchronization problem

D. Memory mapping

☐ Answer: C

20. Deadlocks occur when:

- A. Processes share no resource
- B. No process is blocked
- C. Processes wait for resources held by each other
- D. Memory is full

☐ **Answer: C**

21. Monitors are used in:

- A. File handling
- B. I/O management
- C. Process synchronization
- D. Process creation

☐ **Answer: C**

22. A mutex is:

- A. A pipe
- B. A type of semaphore with binary value
- C. A message queue
- D. An address space

☐ **Answer: B**

23. In UNIX, a signal is:

- A. A variable
- B. A system call
- C. A short message sent to a process
- D. A process

☐ **Answer: C**

24. Which condition is not necessary for a deadlock?

- A. Mutual exclusion
- B. Circular wait
- C. Preemption

D. Hold and wait

☐ Answer: C

25. What is the maximum number of processes that can communicate via pipe?

A. 1

B. 2

C. Unlimited

D. 4

☐ Answer: B

26. Producer-consumer problem is an example of:

A. Memory management

B. Scheduling

C. Interprocess communication and synchronization

D. Paging

☐ Answer: C

27. In message passing, messages are stored in:

A. Registers

B. RAM

C. Message queues

D. Stack

☐ Answer: C

28. A system call used to create a message queue in UNIX is:

A. msgget()

B. msgrcv()

C. send()

D. signal()

☐ Answer: A

29. Blocking send/receive means:

- A. Messages are stored forever
- B. Sender/receiver waits until operation completes
- C. No data is exchanged
- D. Process is terminated

☐ **Answer: B**

30. The main advantage of IPC is:

- A. More RAM
- B. Reduced CPU time
- C. Process collaboration and data sharing
- D. File compression

☐ **Answer: C**

☐ **CUET PG Previous Year Questions – IPC**

Q1. (CUET PG 2023)

Which of the following mechanisms is used for communication between unrelated processes?

- A. Threads
- B. Shared Memory
- C. Pipes
- D. Sockets

☐ **Answer: D**

Q2. (CUET PG 2022)

What is the main drawback of shared memory in IPC?

- A. High speed
- B. Race conditions
- C. Context switching
- D. Compiling issues

☐ **Answer: B**

Q3. (CUET PG 2021)

Which IPC method is best suited for communication over a network?

- A. Pipes
- B. Shared memory
- C. Message queues
- D. Sockets

☐ **Answer: D**

☐ **Operating System – Topic: Concurrency**

1. Concurrency in OS means:

- A. Running multiple processes in parallel
- B. Running one process at a time
- C. Running threads sequentially
- D. Running processes without CPU

☐ **Answer: A**

2. Which of the following is a problem in concurrent systems?

- A. Deadlock
- B. Parallelism
- C. Multithreading
- D. Scheduling

☐ **Answer: A**

3. Critical section is:

- A. Part of memory with critical data
- B. Portion of code where shared resources are accessed
- C. OS kernel code
- D. An interrupt routine

☐ **Answer: B**

4. Which of these is required to solve the critical section problem?

- A. Memory allocation
- B. CPU scheduling
- C. Mutual exclusion
- D. Pipelining

☐ **Answer: C**

5. A solution to critical section problem must satisfy:

- A. Mutual exclusion, progress, bounded waiting
- B. Synchronization, paging, deadlock
- C. Starvation, context switching, deadlock
- D. Virtualization, segmentation, preemption

☐ **Answer: A**

6. Race conditions occur when:

- A. Only one process runs
- B. Shared data is accessed without synchronization
- C. CPU is idle
- D. Deadlock is resolved

☐ **Answer: B**

7. Peterson's algorithm is used for:

- A. Memory management
- B. Process scheduling
- C. Solving critical section problem
- D. File handling

☐ **Answer: C**

8. Which of the following is a synchronization tool?

- A. Paging
- B. Semaphore
- C. Thread

D. Loader

☐ **Answer: B**

9. A semaphore with values 0 or 1 is called:

A. Binary semaphore / Mutex

B. Counting semaphore

C. Real-time semaphore

D. Buffered semaphore

☐ **Answer: A**

10. Deadlock is:

A. A process waiting for memory

B. A set of processes waiting indefinitely for resources

C. Memory overflow

D. CPU overload

☐ **Answer: B**

11. Which condition is NOT required for deadlock?

A. Mutual exclusion

B. Hold and wait

C. Preemption

D. Circular wait

☐ **Answer: C**

12. Starvation in concurrency is:

A. A process terminated by OS

B. A process waiting forever for a resource

C. Process not using memory

D. CPU going idle

☐ **Answer: B**

13. Monitor in concurrency is:

- A. Hardware
- B. Software module for synchronization
- C. Memory checker
- D. CPU regulator

☐ **Answer: B**

14. What is the main idea behind multithreading?

- A. More RAM
- B. Concurrent execution
- C. Disk formatting
- D. Network switching

☐ **Answer: B**

15. The statement wait() in concurrency refers to:

- A. Process sleep
- B. Requesting semaphore
- C. CPU halt
- D. Memory access

☐ **Answer: B**

16. The statement signal() in concurrency refers to:

- A. Process switch
- B. CPU start
- C. Releasing semaphore
- D. System call

☐ **Answer: C**

17. Which of these is NOT a classical synchronization problem?

- A. Producer-consumer
- B. Readers-writers
- C. Dining philosophers

D. Thread pooling

☐ **Answer: D**

18. Which of the following problems illustrates resource sharing in concurrency?

A. Dining philosophers problem

B. Sorting algorithm

C. Binary search

D. File compression

☐ **Answer: A**

19. A concurrent system uses:

A. Shared memory or message passing

B. Paging and segmentation

C. Static allocation

D. Compilation

☐ **Answer: A**

20. The operating system uses context switching to:

A. Run only one process

B. Switch between processes or threads

C. Start the computer

D. Compile code

☐ **Answer: B**

21. Busy waiting occurs when:

A. A process sleeps

B. A process waits using CPU cycles

C. I/O is performed

D. Memory is swapped

☐ **Answer: B**

22. Which language construct helps with concurrency?

- A. Loop
- B. If-else
- C. Thread
- D. Print

☐ **Answer: C**

23. Concurrent processes are:

- A. Always dependent
- B. Unrelated
- C. Running simultaneously or interleaved
- D. Scheduled together

☐ **Answer: C**

24. Which of the following is used to prevent race conditions?

- A. Context switching
- B. Shared memory
- C. Mutual exclusion
- D. Process ID

☐ **Answer: C**

25. Which method provides process synchronization?

- A. Deadlock
- B. Paging
- C. Monitor
- D. Virtual memory

☐ **Answer: C**

26. Critical section problems do not arise in:

- A. Single process systems
- B. Multiprocessor systems
- C. Multithreaded systems

D. Shared memory systems

☐ **Answer: A**

27. Which one is true for counting semaphores?

A. Only used for I/O

B. Can take values greater than 1

C. Used in single-threaded processes

D. Can't be used in synchronization

☐ **Answer: B**

28. The synchronization tool provided by Java is:

A. fork()

B. join()

C. synchronized keyword

D. split()

☐ **Answer: C**

29. In concurrency, atomic operations are:

A. Independent

B. Unbreakable by context switches

C. Related to chemistry

D. Always floating point

☐ **Answer: B**

30. Spinlocks:

A. Use busy waiting

B. Block the process

C. Use semaphores

D. Are software interrupts

☐ **Answer: A**

☐ **CUET PG Previous Year Questions – Concurrency**

Q1. (CUET PG 2022)

Which of the following is used to avoid race conditions in concurrent programming?

- A. Threads
- B. Compiler
- C. Mutex
- D. Paging

☐ **Answer: C**

Q2. (CUET PG 2023)

Critical section problem must satisfy all except:

- A. Mutual exclusion
- B. Progress
- C. Fairness
- D. Bounded waiting

☐ **Answer: C**

Q3. (CUET PG 2021)

Which of these classical problems represents concurrency issues?

- A. Binary Search
- B. Dining Philosophers
- C. Quick Sort
- D. DFS

☐ **Answer: B**

☐ **Operating System – Topic: Synchronization**

1. Process synchronization is required when:

- A. Processes run in batch mode
- B. Processes share resources
- C. Processes do not interact

D. Processes have unique memory

☐ Answer: B

2. The purpose of synchronization is to:

A. Speed up processes

B. Ensure consistent access to shared data

C. Manage file systems

D. Schedule jobs

☐ Answer: B

3. A critical section is:

A. A system call

B. A section where only one process must execute at a time

C. A reserved memory segment

D. Part of the BIOS

☐ Answer: B

4. A solution to the critical section problem must satisfy all EXCEPT:

A. Mutual exclusion

B. Deadlock

C. Progress

D. Bounded waiting

☐ Answer: B

5. Mutual exclusion means:

A. No two processes execute at the same time

B. All processes run simultaneously

C. No need for synchronization

D. Sharing global variables

☐ Answer: A

6. The bounded waiting condition means:

- A. Waiting time must be infinite
- B. Process must eventually get the resource
- C. Starvation is allowed
- D. Priority must be zero

☐ **Answer: B**

7. Peterson's Algorithm solves:

- A. File allocation
- B. Virtual memory
- C. Critical section problem for 2 processes
- D. Deadlock detection

☐ **Answer: C**

8. A mutex is:

- A. A signal
- B. A binary semaphore
- C. A type of memory
- D. Paging algorithm

☐ **Answer: B**

9. A binary semaphore can have values:

- A. 0 and 1 only
- B. 0 to 255
- C. True or False
- D. None of the above

☐ **Answer: A**

10. A counting semaphore can:

- A. Be used only for threads
- B. Take any non-negative value
- C. Block interrupts

D. Only work in kernels

☐ **Answer: B**

11. The wait() operation is used to:

A. Start a process

B. Increase a semaphore

C. Decrease a semaphore or block

D. End execution

☐ **Answer: C**

12. The signal() operation is used to:

A. Send a message

B. Increase the value of a semaphore

C. Switch context

D. Kill a thread

☐ **Answer: B**

13. Spinlocks are:

A. Blocking synchronization tools

B. Hardware based

C. Locks using busy waiting

D. Used only in user mode

☐ **Answer: C**

14. Starvation occurs when:

A. Processes wait indefinitely

B. CPU halts

C. Memory is empty

D. Process terminates

☐ **Answer: A**

15. Deadlock occurs when:

- A. Processes have no shared resources
- B. Processes are not using memory
- C. Each process is waiting for a resource held by another
- D. One process is in infinite loop

☐ **Answer: C**

16. The Dining Philosophers problem is an example of:

- A. Memory leak
- B. Deadlock and synchronization
- C. File locking
- D. I/O control

☐ **Answer: B**

17. In the Readers-Writers problem, readers:

- A. Never access shared memory
- B. Can access simultaneously if no writer is active
- C. Must wait for other readers
- D. Have lower priority

☐ **Answer: B**

18. Monitors are:

- A. Hardware devices
- B. Synchronization tools used in programming languages
- C. Virtual memory managers
- D. Process schedulers

☐ **Answer: B**

19. Java uses which keyword for synchronization?

- A. volatile
- B. atomic
- C. synchronized

D. static

☐ Answer: C

20. Which of the following is NOT used for synchronization?

A. Semaphore

B. Mutex

C. Thread

D. Monitor

☐ Answer: C

21. Deadlock prevention strategies include:

A. Forcing processes to release resources

B. Avoiding mutual exclusion

C. Disabling interrupts

D. Infinite wait

☐ Answer: A

22. Which of the following is not a condition for deadlock?

A. Circular wait

B. Hold and wait

C. Preemption

D. Mutual exclusion

☐ Answer: C

23. Busy waiting is:

A. Process waits using CPU cycles

B. Non-blocking wait

C. Used in interrupt handling

D. Always efficient

☐ Answer: A

24. Synchronization is necessary for:

- A. Thread communication
- B. Compiler optimizations
- C. Disk formatting
- D. GUI rendering

☐ **Answer: A**

25. Mutual exclusion is usually implemented using:

- A. Shared memory
- B. Paging
- C. Locks or semaphores
- D. Registers

☐ **Answer: C**

26. In POSIX, thread synchronization is handled using:

- A. pthread_sync
- B. pthread_join
- C. pthread_mutex
- D. threadsafe

☐ **Answer: C**

27. Which of the following causes starvation?

- A. Fair scheduling
- B. High-priority threads blocking others
- C. FIFO queues
- D. Balanced mutex

☐ **Answer: B**

28. Which synchronization method disables interrupts?

- A. Kernel lock
- B. Hardware lock
- C. Busy waiting

D. Preemption control

☐ **Answer: A**

29. Synchronization is most important in:

A. Single-user OS

B. Batch systems

C. Multiprogramming or multithreading

D. BIOS operations

☐ **Answer: C**

30. Which of the following helps avoid both race conditions and deadlocks?

A. Critical section

B. Context switch

C. Proper use of semaphores

D. Compilers

☐ **Answer: C**

☐ **CUET PG Previous Year Questions – Synchronization**

Q1. (CUET PG 2023)

Which of the following conditions must be satisfied to prevent race conditions?

A. Paging

B. Mutual exclusion

C. Thrashing

D. Interrupts

☐ **Answer: B**

Q2. (CUET PG 2022)

Which of the following solves the critical section problem for 2 processes?

A. Banker's algorithm

B. FIFO

C. Peterson's algorithm

D. Round Robin

□ **Answer: C**

Q3. (CUET PG 2021)

Which of the following is not a synchronization tool?

- A. Monitor
- B. Semaphore
- C. Thread
- D. Mutex

□ **Answer: C**

□ □ **Deadlock: Key Concepts to Revise**

1. **Deadlock:** A situation where a group of processes are blocked because each process is holding a resource and waiting for another.
 2. **Four necessary conditions** for deadlock:
 - o Mutual Exclusion
 - o Hold and Wait
 - o No Preemption
 - o Circular Wait
 3. **Handling strategies:**
 - o Deadlock Prevention
 - o Deadlock Avoidance (e.g., Banker's Algorithm)
 - o Deadlock Detection and Recovery
 - o Deadlock Ignorance (Ostrich policy)
 4. **Safe state:** A state where the system can allocate resources to each process without leading to deadlock.
 5. **Resource Allocation Graph (RAG)** and **Wait-For Graph:** Tools to model and detect deadlock.
-

1. Deadlock occurs when:

- A. Processes run in parallel
- B. Processes hold resources and wait indefinitely
- C. CPU gets overloaded
- D. Cache memory is full

☐ **Answer: B**

☐ *Explanation: Deadlock happens when each process is waiting for a resource held by another, causing a cycle.*

2. Which of the following is NOT a condition for deadlock?

- A. Mutual Exclusion
- B. No Preemption
- C. Safe State
- D. Circular Wait

☐ **Answer: C**

3. In Banker's algorithm, what is checked before granting a request?

- A. System uptime
- B. Safe state
- C. I/O response
- D. CPU speed

☐ **Answer: B**

4. Which graph is used to detect deadlocks in a system?

- A. Resource Allocation Graph
- B. Flowchart
- C. Binary Tree
- D. Process Table

☐ **Answer: A**

5. The condition "Hold and Wait" refers to:

- A. A process holding CPU only
- B. A process requesting all resources at once
- C. A process holding one resource and waiting for others
- D. A process waiting to finish

☐ **Answer: C**

6. What type of resources generally cause deadlock?

- A. Preemptable resources
- B. Non-preemptable resources
- C. Shared resources
- D. External resources

☐ **Answer: B**

7. Which one of the following strategies is used for deadlock avoidance?

- A. Round Robin
- B. Banker's Algorithm
- C. Shortest Job First
- D. Priority Scheduling

☐ **Answer: B**

8. The Ostrich algorithm is used in:

- A. Deadlock detection
- B. Deadlock prevention
- C. Ignoring deadlock
- D. Disk scheduling

☐ **Answer: C**

9. A system is said to be in an unsafe state if:

- A. It cannot guarantee completion of all processes
- B. Deadlock has occurred
- C. All resources are allocated

D. There is no memory

☐ **Answer: A**

10. Which of the following algorithms helps prevent circular wait?

A. LRU

B. FIFO

C. Resource hierarchy

D. Thrashing

☐ **Answer: C**

☐ **CUET PG Previous Year Questions (PYQs)**

Q1. (CUET PG 2022)

Which of the following is NOT a necessary condition for deadlock?

A. Circular Wait

B. No Preemption

C. Hold and Wait

D. Preemptive Scheduling

☐ **Answer: D**

Q2. (CUET PG 2023)

In a deadlock situation, processes are in a state of:

A. Execution

B. Waiting

C. Ready

D. Suspended

☐ **Answer: B**

Q3. (CUET PG 2021)

Which of the following algorithms is used to avoid deadlocks?

A. Round Robin

B. FCFS

C. Banker's Algorithm

D. Shortest Remaining Time First

□ **Answer: C**

□ Quick Recap Table

Condition	Meaning
Mutual Exclusion	At least one resource is non-shareable
Hold and Wait	A process holds one and waits for another
No Preemption	Resource cannot be taken back
Circular Wait	Circular chain of processes waiting on each other

□ CPU Scheduling – Key Concepts

- **CPU Scheduling** decides which process gets the CPU next.
 - **Scheduler Types:**
 - **Long-term:** Decides which jobs are admitted.
 - **Short-term (CPU scheduler):** Picks from ready queue.
 - **Medium-term:** Suspends/resumes processes.
 - **Scheduling Criteria:** CPU utilization, throughput, turnaround time, waiting time, response time.
 - **Scheduling Algorithms:**
 - FCFS (First Come First Serve)
 - SJF (Shortest Job First)
 - RR (Round Robin)
 - Priority Scheduling
 - Multilevel Queue
-

1. Which scheduler decides which process runs on the CPU?

- A. Long-term
- B. Short-term
- C. Medium-term
- D. Disk scheduler

☐ **Answer: B**

2. In FCFS scheduling:

- A. Shortest job gets CPU first
- B. Longest job gets CPU first
- C. Process arriving first gets CPU
- D. Priority determines execution

☐ **Answer: C**

3. SJF scheduling algorithm is optimal because it:

- A. Has low memory usage
- B. Gives minimum average waiting time
- C. Uses minimum CPU cycles
- D. Gives maximum priority to all processes

☐ **Answer: B**

4. In Round Robin, each process is assigned:

- A. Priority
- B. Quantum time
- C. Burst time
- D. Response time

☐ **Answer: B**

5. Which algorithm may lead to starvation?

- A. Round Robin
- B. FCFS
- C. Priority Scheduling

D. Multilevel Feedback Queue

☐ Answer: C

6. Turnaround time is:

A. Completion time - Arrival time

B. Waiting time + Burst time

C. Arrival time - Completion time

D. Response time + Burst time

☐ Answer: A

7. What is the drawback of FCFS?

A. Simple to implement

B. Starvation

C. Convoy effect

D. High throughput

☐ Answer: C

8. In Priority Scheduling, a higher priority process:

A. Waits till FCFS is complete

B. Always runs last

C. Preempts lower priority process

D. Gets less CPU time

☐ Answer: C

9. What determines CPU utilization?

A. Number of processes

B. Amount of I/O

C. Scheduling algorithm

D. CPU busy time

☐ Answer: D

10. Which of the following is non-preemptive?

- A. SJF
- B. Round Robin
- C. Priority (preemptive)
- D. FCFS

☐ **Answer: D**

11. Average waiting time is lowest for:

- A. FCFS
- B. Priority
- C. SJF (non-preemptive)
- D. Round Robin

☐ **Answer: C**

12. Which scheduling algorithm uses time quantum?

- A. FCFS
- B. SJF
- C. Round Robin
- D. Priority

☐ **Answer: C**

13. Response time is the time between:

- A. Arrival and completion
- B. Arrival and first execution
- C. Completion and turnaround
- D. Ready and termination

☐ **Answer: B**

14. In multilevel feedback queue scheduling:

- A. Processes are randomly assigned
- B. Processes are moved between queues
- C. Each queue has same time quantum

D. FIFO is always used

☐ **Answer: B**

15. Which scheduling is ideal for time-sharing systems?

A. Priority Scheduling

B. Round Robin

C. SJF

D. FCFS

☐ **Answer: B**

16. What causes starvation in scheduling?

A. Lack of memory

B. Infinite loop

C. Low priority processes always being skipped

D. High turnaround time

☐ **Answer: C**

17. Which algorithm requires knowledge of burst time?

A. FCFS

B. SJF

C. Round Robin

D. Priority

☐ **Answer: B**

18. Preemptive version of SJF is called:

A. Longest Job First

B. Priority Scheduling

C. Shortest Remaining Time First

D. Multilevel Queue

☐ **Answer: C**

19. The goal of CPU scheduling is to:

- A. Increase response time
- B. Maximize CPU utilization and throughput
- C. Increase turnaround time
- D. Minimize CPU idle time only

☐ **Answer: B**

20. The CPU burst time is:

- A. Time CPU is idle
- B. Time taken for I/O
- C. Time a process uses the CPU continuously
- D. Arrival time

☐ **Answer: C**

☐ **CUET PG Previous Year Questions (CPU Scheduling)**

Q1. (CUET PG 2023)

Which of the following scheduling algorithms can cause starvation?

- A. FCFS
- B. SJF
- C. Round Robin
- D. Priority Scheduling

☐ **Answer: D**

Q2. (CUET PG 2022)

Round Robin scheduling algorithm is most suitable for:

- A. Real-time systems
- B. Time-sharing systems
- C. Batch systems
- D. Embedded systems

☐ **Answer: B**

Q3. (CUET PG 2021)

In which scheduling algorithm can a process be preempted?

- A. FCFS
- B. SJF (non-preemptive)
- C. Priority (preemptive)
- D. All of the above

Answer: C

Summary Table

Algorithm	Preemptive	Starvation Possible	Use Case
FCFS	☐	☐	Simple batch systems
SJF	☐ / ☐	☐ (short jobs only)	Efficient CPU usage
Round Robin	☐	☐	Time-sharing systems
Priority	☐ / ☐	☐	Real-time & embedded

Here is a complete **I/O Scheduling** study and practice pack for your **CUET PG MCA** preparation. It includes:

- ☐ Multiple-choice questions (MCQs) with answers & explanations
 - ☐ CUET PG previous year questions (if available)
 - ☐ Quick theory revision
-

I/O Scheduling - Key Concepts

- **I/O Scheduling** determines the order in which I/O requests are serviced.
- I/O operations are much slower than CPU operations.
- Efficient I/O scheduling helps reduce:
 - o Average seek time
 - o Latency
 - o Overall response time
 - o Disk head movement

Common Disk Scheduling Algorithms:

Algorithm Key Idea

FCFS	First come, first served
SSTF	Shortest Seek Time First
SCAN	Elevator algorithm – moves in one direction
C-SCAN	Circular SCAN – goes in one direction and returns
LOOK	Like SCAN but stops at last request
C-LOOK	Like C-SCAN but stops at last request in one direction

☐ MCQs on I/O Scheduling

1. I/O scheduling is needed to:

- A. Improve CPU usage
- B. Minimize disk arm movement
- C. Reduce RAM usage
- D. Improve thread scheduling

☐ **Answer:** B

2. Which I/O scheduling algorithm services requests in the order they arrive?

- A. SSTF
- B. FCFS
- C. SCAN
- D. LOOK

☐ **Answer:** B

3. SSTF stands for:

- A. Shortest Start Time First
- B. Shortest Seek Time First
- C. Smallest System Transfer First

D. Short System Timing Function

☐ Answer: B

4. Which scheduling causes starvation of faraway requests?

- A. FCFS
- B. SSTF
- C. SCAN
- D. LOOK

☐ Answer: B

5. Which algorithm is also known as the Elevator Algorithm?

- A. FCFS
- B. SSTF
- C. SCAN
- D. LOOK

☐ Answer: C

6. What does SCAN do when it reaches the end of the disk?

- A. Stops
- B. Reverses direction
- C. Goes to the middle
- D. Resets to track 0

☐ Answer: B

7. Which algorithm moves only in one direction and jumps back to the beginning?

- A. LOOK
- B. SCAN
- C. C-SCAN
- D. SSTF

☐ Answer: C

8. What is the main advantage of LOOK over SCAN?

- A. Always resets to zero
- B. Saves head movement by stopping at last request
- C. No starvation
- D. Works only for SSD

☐ **Answer: B**

9. Which of the following is prone to starvation?

- A. FCFS
- B. C-SCAN
- C. SSTF
- D. SCAN

☐ **Answer: C**

10. Which is best for uniformly distributed disk requests?

- A. FCFS
- B. SSTF
- C. SCAN
- D. C-SCAN

☐ **Answer: D**

11. Which I/O scheduling is best for minimizing seek time?

- A. SSTF
- B. FCFS
- C. C-SCAN
- D. Priority Scheduling

☐ **Answer: A**

12. Which scheduling policy ensures fairness to all requests?

- A. SSTF
- B. SCAN
- C. LOOK

D. FCFS

□ **Answer: D**

□ **CUET PG PYQs on I/O Scheduling (If Asked)**

□ As of 2021–2023, **no direct questions on I/O scheduling** have appeared in CUET PG. But similar concepts may be mixed in OS or disk-related questions.

Example (CUET-like):

Q. The disk scheduling algorithm that avoids starvation and ensures fairness is:

A. SSTF

B. LOOK

C. FCFS

D. C-SCAN

□ **Answer: C**

□ **Summary Table: Disk Scheduling Algorithms**

Algorithm Preemptive Starvation Directional

FCFS	□	□	□
SSTF	□	□	□
SCAN	□	□	□
C-SCAN	□	□	□ (single)
LOOK	□	□	□
C-LOOK	□	□	□ (single)

□ **What is Resource Scheduling?**

Resource scheduling in an OS refers to **allocating system resources** (CPU, I/O, memory, devices, etc.) to processes efficiently. The goal is to:

- Maximize system utilization

- Avoid deadlock and starvation
- Ensure fairness
- Maintain high throughput and low waiting time

☐ **Common Resource Scheduling Areas:**

1. **CPU Scheduling** – Assigning CPU to processes
 2. **I/O Scheduling** – Managing I/O device queues
 3. **Disk Scheduling** – Managing disk access
 4. **Memory Scheduling** – Allocating main memory
 5. **Multi-resource scheduling** – Coordinating CPU, memory, I/O together
-

☐ **MCQs on Resource Scheduling**

1. Resource scheduling ensures:

- A. Low I/O speed
- B. Resource conflict
- C. Optimal utilization of resources
- D. Memory fragmentation

☐ **Answer: C**

2. Which resource is typically managed by CPU scheduling?

- A. Disk
- B. I/O devices
- C. CPU time
- D. Virtual memory

☐ **Answer: C**

3. Which of the following is not a resource?

- A. Printer
- B. CPU

- C. File
- D. Instruction

☐ **Answer: D**

4. Starvation in resource scheduling happens when:

- A. A process keeps executing
- B. A low priority process never gets a resource
- C. Resources are shared equally
- D. None of the above

☐ **Answer: B**

5. Deadlock may occur in resource scheduling due to:

- A. Mutual exclusion
- B. Circular wait
- C. Hold and wait
- D. All of the above

☐ **Answer: D**

6. Which of the following helps in avoiding deadlocks?

- A. Resource reservation
- B. Mutual exclusion
- C. Circular wait
- D. Priority scheduling

☐ **Answer: A**

7. Banker's algorithm is used for:

- A. Memory management
- B. File system
- C. Deadlock avoidance in resource allocation
- D. CPU scheduling

☐ **Answer: C**

8. When multiple resources are needed by a process, the safest strategy is:

- A. Request one by one
- B. Request all at once
- C. Use FCFS
- D. Preempt high-priority resource

☐ **Answer: B**

9. Which resource cannot be preempted easily?

- A. CPU
- B. Memory
- C. Printer
- D. Network

☐ **Answer: C**

10. Round Robin resource scheduling is used in:

- A. Real-time system
- B. Time-sharing system
- C. Batch processing
- D. Distributed systems

☐ **Answer: B**

11. In multilevel feedback queues:

- A. One resource queue is used
- B. No priority is given
- C. Multiple queues for resource types
- D. Starvation always occurs

☐ **Answer: C**

12. To prevent starvation, we can use:

- A. Static priorities
- B. Aging (gradually increase priority)
- C. Mutual exclusion

D. Preemption

☐ **Answer: B**

13. A resource is in use by one process and cannot be used by another simultaneously. This is called:

A. Preemption

B. Mutual exclusion

C. Shared mode

D. Interrupt

☐ **Answer: B**

14. If all processes request and hold resources in circular fashion, it leads to:

A. Synchronization

B. Thrashing

C. Starvation

D. Deadlock

☐ **Answer: D**

15. Which is not a strategy for resource scheduling?

A. Preemption

B. Deadlock detection

C. Lazy loading

D. Wait-die scheme

☐ **Answer: C**

☐ **CUET PG PYQ-style Sample Questions**

(Note: "Resource scheduling" may be indirectly asked under deadlock, CPU scheduling, or system performance.)

Q1. (CUET Style)

A system has 3 printers. P1 holds 1 and requests 1, P2 holds 1 and requests 1. If P3 also requests 1, what must be done to avoid deadlock?

- A. Allow P3 to wait
- B. Preempt printer from P1
- C. Deny request by P3
- D. Allow P2 to finish

☐ **Answer: C**

Q2.

Which of the following conditions is not necessary for deadlock in resource scheduling?

- A. Circular wait
- B. Mutual exclusion
- C. Preemption
- D. Hold and wait

☐ **Answer: C**

Q3.

Which algorithm ensures **safe resource allocation**?

- A. FIFO
- B. FCFS
- C. Banker's algorithm
- D. Look scheduling

☐ **Answer: C**

☐ **Summary - Resource Scheduling Concepts**

Concept	Description
Starvation	Low-priority process never gets resource
Deadlock	Processes wait forever for each other's resources
Preemption	Taking a resource forcibly from a process
Mutual Exclusion	Only one process can use resource at a time
Banker's Algorithm	Checks for safe state before granting request

Concept	Description
Aging	Increases priority of waiting processes

☐ Topic: Deadlock & Scheduling Algorithms

☐ SECTION 1: Deadlock – MCQs (with Answers)

1. Which of the following is not a necessary condition for deadlock?

- A. Mutual exclusion
- B. Circular wait
- C. Preemption
- D. Hold and wait

☐ **Answer: C**

2. What is the safe state in deadlock avoidance?

- A. State where deadlock is certain
- B. State where system can allocate resources without entering deadlock
- C. State where resources are never allocated
- D. System halts execution

☐ **Answer: B**

3. Which algorithm is used to avoid deadlock?

- A. Dijkstra's algorithm
- B. Banker's algorithm
- C. FIFO algorithm
- D. Round Robin algorithm

☐ **Answer: B**

4. What does the wait-for graph detect?

- A. Starvation
- B. Thrashing
- C. Deadlock
- D. Page faults

☐ **Answer: C**

5. Which is a deadlock prevention technique?

- A. Allow circular wait
- B. Request all resources at once
- C. Allow hold and wait
- D. Allow preemption always

☐ **Answer: B**

☐ **SECTION 2: CPU Scheduling – MCQs (with Answers)**

6. Which scheduling algorithm gives minimum average waiting time for a given set of processes?

- A. FCFS
- B. SJF
- C. Round Robin
- D. Priority Scheduling

☐ **Answer: B**

7. In Round Robin, what does the time quantum refer to?

- A. Process arrival time
- B. Waiting time
- C. CPU burst time
- D. Time a process is allowed to run before switching

☐ **Answer: D**

8. Which scheduling algorithm can cause starvation?

- A. SJF
- B. FCFS
- C. Round Robin
- D. Multilevel Queue

☐ **Answer: A**

9. Which of these is a non-preemptive scheduling algorithm?

- A. FCFS
- B. Round Robin
- C. SRTF
- D. Priority (preemptive)

☐ **Answer: A**

10. Aging is used to solve the problem of:

- A. Thrashing
- B. Deadlock
- C. Starvation
- D. Paging

☐ **Answer: C**

☐ **CUET PG PYQs & Practice**

Q1. (CUET PG 2023)

Which of the following is a necessary condition for deadlock to occur?

- A. Preemption
- B. Context Switching
- C. Hold and Wait
- D. Paging

☐ **Answer: C**

Q2. (CUET PG 2022)

Round Robin scheduling algorithm is suitable for:

- A. Batch processing systems
- B. Real-time systems
- C. Time-sharing systems
- D. Distributed systems

☐ **Answer: C**

Q3. (CUET PG 2021)

In which condition is deadlock most likely to occur?

- A. Processes are completed
- B. Circular wait is not allowed
- C. Resources are shared and limited
- D. Page fault occurs

☐ **Answer: C**

Q4. (CUET PG Mock Test)

In Banker's Algorithm, what does the system check before granting a resource?

- A. Whether the CPU is idle
- B. Whether the system remains in safe state
- C. Whether the page table is full
- D. Whether the process is in critical section

☐ **Answer: B**

☐ **Summary Table**

Deadlock Condition Must be Prevented?

Mutual Exclusion	No (some resources must be exclusive)
Hold and Wait	Yes
No Preemption	Yes
Circular Wait	Yes

Scheduling Algorithm Preemptive Starvation Possible Type

FCFS	☐ No	No	Non-preemptive
SJF	☐ / ☐	Yes	Depends
Round Robin	☐ Yes	No	Preemptive
Priority Scheduling	☐ / ☐	Yes (without aging)	Both

11. Deadlock can be completely avoided by:

- A. Disabling interrupts
- B. Allocating resources one at a time
- C. Preventing circular wait
- D. Allowing mutual exclusion

☐ **Answer: C**

12. What does Banker's Algorithm require to work?

- A. Priority of processes
- B. Maximum resource requirement of each process
- C. Arrival time of process
- D. Page reference string

☐ **Answer: B**

13. Which strategy is not used for handling deadlocks?

- A. Prevention
- B. Detection and recovery
- C. Ignorance
- D. Compaction

☐ **Answer: D**

14. What does "No preemption" mean in deadlock conditions?

- A. CPU can't be taken back
- B. Resources can't be forcibly taken back
- C. OS can terminate process
- D. No interrupt allowed

☐ **Answer: B**

15. Which tool is used to detect deadlocks in a system with single instance of resources?

- A. Wait-for graph
- B. Resource-allocation graph
- C. Deadlock detection matrix
- D. Page table

☐ **Answer: A**

☐ **More MCQs: CPU Scheduling Algorithms**

16. In SJF, if arrival times vary, what version of SJF must be used?

- A. Preemptive (SRTF)
- B. Round Robin
- C. FCFS
- D. Priority

☐ **Answer: A**

17. Which algorithm is best suited for a real-time operating system?

- A. FCFS
- B. Round Robin
- C. Priority Scheduling
- D. SJF

☐ **Answer: C**

18. Average turnaround time is calculated as:

- A. Finish Time – Arrival Time
- B. Waiting Time + CPU Burst Time

- C. Start Time – Arrival Time
- D. Arrival Time – Finish Time

☐ **Answer: A**

19. Which is preemptive?

- A. FCFS
- B. Non-preemptive priority
- C. SRTF
- D. SJF

☐ **Answer: C**

20. In Round Robin with a very large time quantum, it behaves like:

- A. SJF
- B. FCFS
- C. Priority
- D. Random

☐ **Answer: B**

Q1. What is the main cause of starvation in scheduling?

- A. Lack of CPU time
- B. High-priority processes keep getting executed
- C. Page faults
- D. Infinite loops

Q2. In Banker's Algorithm, which state ensures no deadlock?

- A. Unsafe
- B. Critical
- C. Ready
- D. Safe

Q3. What is the turn-around time formula?

- A. Start time – Arrival time
- B. Finish time – Arrival time

- C. Waiting time + Arrival time
- D. Burst time – Waiting time

Q4. In which algorithm can starvation occur if priorities are not adjusted?

- A. Round Robin
- B. FCFS
- C. SJF
- D. Priority

Q5. Deadlock detection is applicable when system allows:

- A. Prevention
- B. Circular wait
- C. Resource over-allocation
- D. Preemption

Q6. Which scheduling algorithm reduces average waiting time most effectively?

- A. FCFS
- B. SJF
- C. Priority
- D. Round Robin

Q7. Circular wait can be avoided by:

- A. Assigning priorities to resources
- B. Using FCFS
- C. Swapping
- D. Preemption

Q8. Which is the best choice for time-sharing systems?

- A. Round Robin
- B. SJF
- C. FCFS
- D. Priority

Q9. The condition where a process holds one resource and waits for another is called:

- A. Mutual Exclusion
- B. Hold and Wait
- C. Circular wait
- D. Starvation

Q10. If a process is continuously denied access to resources, it leads to:

- A. Thrashing
- B. Starvation

- C. Paging
 - D. Deadlock
-

☐ Mock Test Answers:

1. B
 2. D
 3. B
 4. D
 5. C
 6. B
 7. A
 8. A
 9. B
 10. B
-

☐ Part 1: MCQs with Answers – Banker's Algorithm

Q1. What is the main goal of Banker's Algorithm?

- A. Deadlock detection
- B. Deadlock prevention
- C. Deadlock avoidance
- D. CPU scheduling

☐ Answer: C

Q2. Which of the following is *not* a valid matrix in Banker's Algorithm?

- A. Max
- B. Allocation

C. Available

D. Transfer

☐ **Answer: D**

Q3. The Need matrix is calculated using:

A. Allocation – Max

B. Max – Available

C. Max – Allocation

D. Allocation + Max

☐ **Answer: C**

Q4. If the system is in a safe state, it means:

A. No process can proceed

B. System is deadlocked

C. There exists a safe sequence

D. Resources are underutilized

☐ **Answer: C**

Q5. If a process requests more than its maximum declared need, the system should:

A. Grant the request

B. Wait until available

C. Treat it as an error

D. Suspend the process

☐ **Answer: C**

Q6. What does the Available vector in Banker's Algorithm represent?

A. Max resource needs

B. Total resources

C. Resources currently free

D. Allocated resources

☐ **Answer: C**

Q7. What happens if a request passes the safety check?

- A. Deadlock occurs
- B. Request is denied
- C. Resources are allocated
- D. System shuts down

☐ **Answer: C**

Q8. Which of the following is a valid safety condition in Banker's Algorithm?

- A. $\text{Need} > \text{Work}$
- B. $\text{Need} \leq \text{Work}$
- C. $\text{Allocation} = \text{Need}$
- D. $\text{Max} = \text{Available}$

☐ **Answer: B**

Q9. Banker's Algorithm was originally designed for:

- A. Batch systems
- B. Multiprocessing systems
- C. Single-user systems
- D. Banking systems

☐ **Answer: D**

Q10. Which resource allocation state is not possible in a system using Banker's Algorithm?

- A. Unsafe
- B. Deadlocked
- C. Safe
- D. Partial allocation

☐ **Answer: B**

Q11. If a system is in an unsafe state, it means:

- A. Deadlock is definitely present
- B. Deadlock may occur
- C. System is idle

D. System is fully utilized

☐ **Answer: B**

Q12. A system has 3 resource types and 5 processes. What is the max number of resource instances needed to ensure deadlock-free allocation?

- A. Sum of max need of all processes
- B. One less than total resources
- C. One more than max allocation
- D. Banker's Algorithm decides based on need and availability

☐ **Answer: D**

Q13. Can the Banker's Algorithm handle dynamic processes (i.e., new processes added at runtime)?

- A. Yes
- B. No
- C. Only if they request zero resources
- D. Only if memory allows

☐ **Answer: B**

Q14. Which of the following steps is not a part of Banker's Algorithm?

- A. Checking Request $\leq$ Need
- B. Checking Request $\leq$ Available
- C. Granting request and testing for safe state
- D. Deallocating unused memory

☐ **Answer: D**

Q15. Which data structure is most helpful in the safety algorithm used in Banker's Algorithm?

- A. Queue
- B. Stack
- C. Vector
- D. Graph

☐ **Answer: C**

□ Part 2: PYQs (CUET + Other MCA Exams)

PYQ 1:

You are given:

- Available: [3, 3, 2]
- Allocation:

P0: [0, 1, 0]

P1: [2, 0, 0]

P2: [3, 0, 2]

P3: [2, 1, 1]

P4: [0, 0, 2]

- Max:

P0: [7, 5, 3]

P1: [3, 2, 2]

P2: [9, 0, 2]

P3: [2, 2, 2]

P4: [4, 3, 3]

Is the system in a safe state?

□ **Answer:** Yes

□ **Safe sequence:** P1 → P3 → P4 → P0 → P2

PYQ 2:

If Available = [1,2,1], and Request = [1,1,1], but Need = [2,2,1], what happens?

- A. Request granted
- B. Denied: Request > Need
- C. Denied: Request > Available

D. Both B and C

☐ **Answer:** D

Great! Here's a **set of numerical MCQs on Banker's Algorithm** for deep CUET PG MCA practice. These are ideal for mastering allocation, need calculation, and safe state checks.

☐ **Numerical-Based MCQs – Banker's Algorithm**

Q1.

Available: [3, 3, 2]

Allocation:

P0: [0, 1, 0]

P1: [2, 0, 0]

P2: [3, 0, 2]

P3: [2, 1, 1]

P4: [0, 0, 2]

Max:

P0: [7, 5, 3]

P1: [3, 2, 2]

P2: [9, 0, 2]

P3: [2, 2, 2]

P4: [4, 3, 3]

Is the system in a safe state?

A. Yes, safe sequence exists

B. No, deadlock occurs

C. Can't determine

D. System is idle

☐ **Answer:** A (Safe sequence: P1 → P3 → P4 → P0 → P2)

Q2.

Process P2 requests [2, 0, 0]

Need(P2) = [6, 0, 0]

Available = [3, 2, 1]

Should the system grant the request?

A. Yes, request $\leq$ Need and $\leq$ Available

B. No, request > Available

C. No, request > Need

D. Not enough info

☐ **Answer: A**

Q3.

You have:

Max = [7, 5, 3], Allocation = [0, 1, 0]

What is the Need?

A. [7, 4, 3]

B. [0, 0, 0]

C. [7, 6, 3]

D. [1, 5, 3]

☐ **Answer: A**

(Need = Max - Allocation = [7-0, 5-1, 3-0] = [7, 4, 3])

Q4.

Available = [1, 2, 1]

Request = [1, 1, 1]

Need = [2, 2, 1]

What happens?

A. Request granted

B. Denied: Request > Need

C. Denied: Request > Available

D. Both B and C

☐ **Answer: D**

Q5.

Available: [3, 2, 2]

Request(P1): [1, 2, 1]

Need(P1): [1, 2, 2]

Allocation(P1): [1, 0, 0]

What is the new Allocation and Available if request is granted?

A. Allocation: [2, 2, 1]; Available: [2, 0, 1]

B. Allocation: [2, 2, 1]; Available: [1, 0, 1]

C. Allocation: [1, 0, 0]; Available: [3, 2, 2]

D. Allocation: [0, 0, 0]; Available: [4, 2, 2]

☐ **Answer: A**

☐ **Summary Cheat Notes**

☐ **Key Formulas:**

- Need = Max – Allocation
- Safe sequence exists if all processes can complete with **Available $\geq$ Need**

☐ **Safety Check Order:**

1. Check: Request $\leq$ Need
 2. Check: Request $\leq$ Available
 3. Pretend to allocate → check if system remains in **safe state**
 4. If yes → **Grant**; If no → **Deny**
-

☐ **Topic: Memory Management & Virtual Memory – MCQs (30 Questions)**

1. What is the primary function of memory management?

- A. Control CPU usage
- B. Track I/O operations
- C. Manage allocation and deallocation of memory
- D. Manage user inputs

☐ **Answer: C**

2. Paging helps in:

- A. Reducing page faults
- B. Eliminating external fragmentation
- C. Reducing context switches
- D. Allocating I/O devices

☐ **Answer: B**

3. Which causes internal fragmentation?

- A. Paging
- B. Segmentation
- C. Contiguous allocation
- D. Swapping

☐ **Answer: A**

4. In segmentation, each segment has:

- A. Same size
- B. Fixed size
- C. Logical name and length
- D. Frame size

☐ **Answer: C**

5. What is a page frame?

- A. Logical unit in memory
- B. Physical block of memory
- C. Segment block
- D. Disk partition

☐ **Answer: B**

6. A page fault occurs when:

- A. Page is in memory
- B. Page is invalid

- C. Page is not in memory
- D. Page is in TLB

☐ **Answer: C**

7. The page table stores:

- A. Virtual addresses
- B. Process IDs
- C. Mapping of logical to physical addresses
- D. CPU scheduling info

☐ **Answer: C**

8. Which algorithm can lead to Belady's Anomaly?

- A. LRU
- B. Optimal
- C. FIFO
- D. MRU

☐ **Answer: C**

9. Which is not a page replacement algorithm?

- A. LRU
- B. FIFO
- C. Optimal
- D. Best fit

☐ **Answer: D**

10. Which technique allows programs larger than RAM to execute?

- A. Compaction
- B. Virtual Memory
- C. Cache Memory
- D. Swapping

☐ **Answer: B**

11. Thrashing occurs when:

- A. RAM is upgraded
- B. Page fault rate is very high
- C. Process has low priority
- D. Segmentation fails

☐ **Answer: B**

12. The working set of a process is:

- A. RAM assigned to OS
- B. Set of active processes
- C. Set of pages used recently
- D. Set of all page tables

☐ **Answer: C**

13. Page size is 1 KB, Logical Address = 2049.

Page Number = ?

- A. 1
- B. 2
- C. 3
- D. 4

☐ **Answer: B**

14. What is the offset for address 2049 with 1 KB page?

- A. 1024
- B. 1
- C. 25
- D. 1025

☐ **Answer: B**

15. Segmentation avoids:

- A. Internal fragmentation
- B. External fragmentation

- C. Thrashing
- D. Page faults

☐ **Answer: A**

16. Which OS scheme requires translation lookaside buffer (TLB)?

- A. Segmentation
- B. Paging
- C. Contiguous allocation
- D. Swapping

☐ **Answer: B**

17. Which memory allocation method is fastest?

- A. Paging
- B. Segmentation
- C. Contiguous allocation
- D. Swapping

☐ **Answer: C**

18. Logical address space = 16 bits, page size = 1KB.

How many pages?

- A. 64
- B. 128
- C. 256
- D. 16

☐ **Answer: C**

19. The Optimal page replacement algorithm:

- A. Uses recent references
- B. Is not implementable
- C. Replaces oldest page
- D. Replaces smallest page

☐ **Answer: B**

20. Page Replacement Algorithm with least faults:

- A. FIFO
- B. Optimal
- C. LRU
- D. Random

☐ **Answer: B**

21. Physical Address = Frame No. + _____

- A. Segment
- B. Offset
- C. TLB
- D. Page

☐ **Answer: B**

22. In contiguous allocation, the problem is:

- A. External fragmentation
- B. Page fault
- C. Internal fragmentation
- D. TLB miss

☐ **Answer: A**

23. The dirty bit in a page table indicates:

- A. Page is in memory
- B. Page is modified
- C. Page is corrupted
- D. Page is swapped

☐ **Answer: B**

24. Which memory is fastest?

- A. Main Memory
- B. Cache

- C. Virtual Memory
- D. Disk

☐ **Answer: B**

25. Which best describes demand paging?

- A. All pages loaded initially
- B. Only required pages loaded
- C. No paging used
- D. Segments used instead

☐ **Answer: B**

26. Virtual memory increases:

- A. Execution speed
- B. Address space
- C. CPU scheduling
- D. Cache size

☐ **Answer: B**

27. LRU stands for:

- A. Least Reused Unit
- B. Least Recently Used
- C. Last Recently Used
- D. Largest Reserved Unit

☐ **Answer: B**

28. In segmentation, which register holds segment base?

- A. Stack pointer
- B. Program counter
- C. Segment register
- D. TLB

☐ **Answer: C**

29. A large page size leads to:

- A. More page faults
- B. More internal fragmentation
- C. Less memory
- D. More external fragmentation

☐ **Answer: B**

30. What is the main goal of swapping?

- A. Increase CPU speed
- B. Replace CPU
- C. Free up RAM
- D. Clean cache

☐ **Answer: C**

Topic: File Systems (CUET PG + Practice)

Total Questions: 30 (Mix of PYQs + Practice)

☐ **Answer key included below**

1. Which of the following is not a file attribute? (CUET PG PYQ)

- A. Name
- B. Type
- C. Size
- D. Process ID

☐ **Answer: D**

2. A file system provides which of the following?

- A. Communication
- B. Memory management
- C. Data management and access
- D. Task scheduling

☐ **Answer: C**

3. In a hierarchical file system, the topmost directory is called:

- A. Home directory
- B. Parent directory
- C. Root directory
- D. Sub-directory

☐ **Answer: C**

4. File extension helps in:

- A. Security
- B. Identifying file type
- C. File compression
- D. Reducing file size

☐ **Answer: B**

5. Which directory structure allows users to share files easily?

- A. Single level
- B. Two level
- C. Tree
- D. Acyclic graph

☐ **Answer: D**

6. In a UNIX system, which command is used to change directory?

- A. cd
- B. ls
- C. mv
- D. pwd

☐ **Answer: A**

7. Which of the following is not a valid file access method?

- A. Sequential
- B. Direct
- C. Indexed

D. Contiguous

☐ **Answer: D**

8. Which file allocation method suffers the most from external fragmentation? (CUET PG PYQ)

A. Linked

B. Indexed

C. Contiguous

D. Hashed

☐ **Answer: C**

9. Which is the fastest file access method?

A. Sequential

B. Indexed

C. Random

D. Direct

☐ **Answer: D**

10. Which allocation method uses pointers in each block?

A. Contiguous

B. Linked

C. Indexed

D. Hashed

☐ **Answer: B**

11. FAT in file systems stands for:

A. File Access Table

B. File Allocation Table

C. File Assignment Table

D. File Architecture Table

☐ **Answer: B**

12. Which of the following is an advantage of indexed allocation?

- A. No external fragmentation
- B. Fastest access
- C. No internal fragmentation
- D. No need for pointers

☐ **Answer: A**

13. Which file system is default in most Linux OS?

- A. NTFS
- B. FAT32
- C. EXT4
- D. HFS

☐ **Answer: C**

14. What is a block in file system context?

- A. Process
- B. CPU unit
- C. Fixed-size unit of data storage
- D. Instruction set

☐ **Answer: C**

15. Which of the following allows non-contiguous allocation? (CUET PG PYQ)

- A. Contiguous
- B. Linked
- C. Sequential
- D. Tree

☐ **Answer: B**

16. Which command is used to display file permissions in Unix?

- A. chmod
- B. ls -l
- C. mkdir

D. ps

☐ Answer: B

17. UNIX uses which file system by default?

A. FAT

B. HFS

C. EXT

D. NTFS

☐ Answer: C

18. What happens in internal fragmentation?

A. Wasted space between blocks

B. Wasted space within allocated block

C. Data loss

D. Security breach

☐ Answer: B

19. The technique of assigning a disk block to more than one file is:

A. Clustering

B. Sharing

C. Aliasing

D. Linking

☐ Answer: C

20. Directory structure used by MS-DOS:

A. Single level

B. Two level

C. Tree

D. Acyclic graph

☐ Answer: C

21. Which of the following systems uses NTFS?

- A. Linux
- B. Unix
- C. Windows
- D. MacOS

☐ **Answer: C**

22. What is the role of the inode in a file system?

- A. Store user data
- B. Store file name
- C. Store file metadata (size, permissions, etc.)
- D. Handle interrupts

☐ **Answer: C**

23. Maximum file size in FAT32 system is:

- A. 2 GB
- B. 4 GB
- C. 8 GB
- D. 16 GB

☐ **Answer: B**

24. How many file allocation methods are there generally?

- A. 2
- B. 3
- C. 4
- D. 5

☐ **Answer: B (Contiguous, Linked, Indexed)**

25. File permissions in Linux include:

- A. Read, Write
- B. Read, Execute
- C. Write, Execute

D. Read, Write, Execute

☐ Answer: D

26. Which file access method is best for large files with frequent updates?

A. Contiguous

B. Linked

C. Indexed

D. Tree

☐ Answer: C

27. Which component manages file security and access control?

A. Kernel

B. File Table

C. OS Scheduler

D. File Manager

☐ Answer: D

28. Which of the following is not stored in a directory entry?

A. File name

B. File size

C. File path

D. File location on disk

☐ Answer: C

29. To remove a file in UNIX, you use:

A. del

B. rm

C. erase

D. remove

☐ Answer: B

30. Files that are read-only have what permission disabled?

- A. Execute
- B. Write
- C. Read
- D. Run

☐ **Answer: B**

☐ **Topic: I/O Systems**

Total MCQs: 30

☐ **Answer key provided at the end**

1. What is the full form of I/O in operating systems?

- A. Input/Output
- B. Internal/Output
- C. Inward/Output
- D. Input/Operate

☐ **Answer: A**

2. The main function of an I/O system is to:

- A. Manage memory
- B. Manage user interface
- C. Manage devices and data exchange
- D. Manage CPU scheduling

☐ **Answer: C**

3. Which of the following is a block device?

- A. Keyboard
- B. Printer
- C. Hard disk
- D. Mouse

☐ **Answer: C**

4. Which of the following is a character device?

- A. USB drive
- B. Hard disk
- C. Scanner
- D. SSD

☐ **Answer: C**

5. Spooling is used for: (CUET PYQ)

- A. Disk scheduling
- B. Memory allocation
- C. Device independence
- D. Managing I/O requests like printing

☐ **Answer: D**

6. What does "device driver" do in an I/O system?

- A. Drives memory to cache
- B. Converts user instructions into hardware commands
- C. Schedules CPU
- D. Manages interrupts

☐ **Answer: B**

7. DMA stands for:

- A. Direct Machine Access
- B. Digital Memory Access
- C. Direct Memory Access
- D. Direct Module Access

☐ **Answer: C**

8. The purpose of DMA is to:

- A. Transfer data without CPU involvement
- B. Manage processes
- C. Manage files

D. Schedule interrupts

☐ **Answer: A**

9. Which component signals the CPU that an I/O device needs attention?

A. Register

B. Memory

C. Interrupt

D. Cache

☐ **Answer: C**

10. Buffering in I/O systems is used to:

A. Increase CPU speed

B. Reduce memory use

C. Store data temporarily during transfers

D. Prevent file access

☐ **Answer: C**

11. Which is not an I/O technique?

A. Programmed I/O

B. Interrupt-driven I/O

C. Memory-mapped I/O

D. Paged I/O

☐ **Answer: D**

12. In programmed I/O, CPU:

A. Delegates I/O to DMA

B. Waits while the I/O completes

C. Uses polling

D. Goes into sleep mode

☐ **Answer: B**

13. Which I/O method is most efficient for large data transfers?

- A. Programmed I/O
- B. Interrupt-driven I/O
- C. DMA
- D. Memory-mapped I/O

☐ **Answer: C**

14. Which of the following helps in asynchronous communication?

- A. Polling
- B. Interrupts
- C. Buffering
- D. Caching

☐ **Answer: B**

15. The technique that allows a device to access memory without involving the CPU is:

- A. Spooling
- B. DMA
- C. Paging
- D. Buffering

☐ **Answer: B**

16. Which of the following is not a typical I/O device?

- A. Monitor
- B. CPU
- C. Printer
- D. Scanner

☐ **Answer: B**

17. Which I/O technique consumes CPU time in polling loop?

- A. DMA
- B. Programmed I/O
- C. Interrupt-driven I/O

D. Spooling

☐ Answer: B

18. Which of the following is true about spooling? (CUET PG PYQ)

- A. Spooling reduces CPU load
- B. Spooling is only for output devices
- C. Spooling uses disk as temporary storage
- D. Spooling is same as buffering

☐ Answer: C

19. Which is an example of a non-blocking I/O?

- A. Interrupt-driven I/O
- B. Programmed I/O
- C. Polling
- D. Blocking I/O

☐ Answer: A

20. Memory-mapped I/O maps I/O devices to:

- A. CPU
- B. RAM
- C. Address space
- D. Cache

☐ Answer: C

21. What is the purpose of an I/O scheduler?

- A. Schedule processes
- B. Decide the order of I/O requests
- C. Schedule memory blocks
- D. Manage cache

☐ Answer: B

22. In interrupt-driven I/O, when is the CPU notified?

- A. At the start of I/O
- B. After polling
- C. When I/O is complete
- D. When the user logs in

☐ **Answer: C**

23. A device controller connects:

- A. CPU and ALU
- B. User and system
- C. OS and I/O device
- D. Memory and file system

☐ **Answer: C**

24. Which statement about I/O devices is false?

- A. Keyboard is a character device
- B. Disk is a block device
- C. All I/O devices are character devices
- D. Printer is a character device

☐ **Answer: C**

25. Which mechanism allows simultaneous I/O operations and CPU execution?

- A. Spooling
- B. Polling
- C. DMA
- D. Paging

☐ **Answer: C**

26. Which of these can lead to CPU busy waiting?

- A. DMA
- B. Spooling
- C. Polling

D. Interrupts

☐ **Answer: C**

27. Which is NOT an advantage of interrupt-driven I/O?

A. Efficient CPU usage

B. Real-time handling

C. No need to poll continuously

D. High latency

☐ **Answer: D**

28. Which file contains information about connected I/O devices in Linux?

A. /home

B. /dev

C. /sys

D. /etc

☐ **Answer: B**

29. A scanner is an example of which device type?

A. Block

B. File

C. Character

D. Network

☐ **Answer: C**

30. Which of the following reduces I/O bottlenecks?

A. Threading

B. Buffering and spooling

C. Virtual memory

D. Paging

☐ **Answer: B**

☐ **Topic: DOS (Disk Operating System)**

Total MCQs: 30

☐ **Answer key at the end**

1. What does DOS stand for?

- A. Digital Operating System
- B. Disk Operating System
- C. Data Operating Software
- D. Disk Operating Software

☐ **Answer: B**

2. DOS is a:

- A. Graphical User Interface
- B. Command Line Interface
- C. Multi-user OS
- D. Real-time OS

☐ **Answer: B**

3. The file extension .exe in DOS stands for:

- A. Execution
- B. Executable
- C. Example
- D. Expansion

☐ **Answer: B**

4. Which command is used to list files in a directory in DOS?

- A. LIST
- B. SHOW
- C. DIR
- D. LST

☐ **Answer: C**

5. In DOS, CD command is used to:

- A. Copy a directory
- B. Clear disk
- C. Change directory
- D. Create directory

☐ **Answer: C**

6. The command to create a directory in DOS is:

- A. CD
- B. MD
- C. RD
- D. DIR

☐ **Answer: B**

7. The command COPY CON FILE.TXT does what?

- A. Copies file.txt
- B. Connects file.txt
- C. Creates a file and lets you type content
- D. Copies contents to CON

☐ **Answer: C**

8. Which command is used to remove a file in DOS?

- A. DEL
- B. ERASE
- C. REMOVE
- D. Both A and B

☐ **Answer: D**

9. DOS is a:

- A. Multi-tasking OS
- B. Single-tasking OS
- C. Distributed OS

D. Network OS

☐ **Answer: B**

10. Which command shows the version of DOS?

A. VERSION

B. VER

C. VRSN

D. DOSVER

☐ **Answer: B**

11. What type of operating system is DOS?

A. GUI based

B. Multitasking

C. Menu-driven

D. Command-line

☐ **Answer: D**

12. The command FORMAT A: does what?

A. Formats hard disk

B. Formats disk in drive A

C. Deletes all files

D. Restarts DOS

☐ **Answer: B**

13. In DOS, AUTOEXEC.BAT is used for:

A. Auto backup

B. Auto formatting

C. Startup commands

D. BIOS settings

☐ **Answer: C**

14. CONFIG.SYS file in DOS is used to:

- A. Set system date
- B. Load drivers and configurations
- C. Store user commands
- D. Copy system files

☐ **Answer: B**

15. Wildcard * in DOS is used to:

- A. Replace one character
- B. Delete all files
- C. Replace any number of characters
- D. Match digits

☐ **Answer: C**

16. What does RD command do?

- A. Removes directory
- B. Reads file
- C. Restores file
- D. Renames directory

☐ **Answer: A**

17. The command to rename a file in DOS is:

- A. RN
- B. NAME
- C. REN
- D. RENAME

☐ **Answer: C**

18. .SYS files in DOS are:

- A. Games
- B. System files
- C. Text files

D. Executable files

☐ **Answer: B**

19. The DOS command to see current working directory is:

A. CD

B. DIR

C. CHKDIR

D. PATH

☐ **Answer: A**

20. Which DOS command will display the contents of a file?

A. DISPLAY

B. VIEW

C. TYPE

D. LIST

☐ **Answer: C**

21. In DOS, internal commands are:

A. Stored in separate files

B. Present in COMMAND.COM

C. Saved in CONFIG.SYS

D. Only run during boot

☐ **Answer: B**

22. What is the maximum filename length in DOS (excluding extension)?

A. 8 characters

B. 16 characters

C. 32 characters

D. 255 characters

☐ **Answer: A**

23. In DOS, file names follow the convention:

- A. 8.3
- B. 6.3
- C. 10.2
- D. 8.8

☐ **Answer: A**

24. The command TREE in DOS is used to:

- A. Display files in tree format
- B. Display folder hierarchy
- C. Format a directory
- D. Create a tree directory

☐ **Answer: B**

25. What does CLS do in DOS?

- A. Clears screen
- B. Closes file
- C. Clears storage
- D. Closes shell

☐ **Answer: A**

26. The command CHKDSK is used for:

- A. Checking keyboard
- B. Checking disk errors
- C. Checking internet
- D. Checking file types

☐ **Answer: B**

27. DOS was introduced in which year?

- A. 1980
- B. 1981
- C. 1983

D. 1985

☐ Answer: B

28. DOS was initially developed by:

A. Microsoft

B. Apple

C. IBM

D. Oracle

☐ Answer: A

29. VOL command is used to:

A. Show free space

B. Show volume label

C. Show file size

D. Show version

☐ Answer: B

30. DOS commands can be classified as:

A. Internal & External

B. Static & Dynamic

C. Real & Virtual

D. Online & Offline

☐ Answer: A

Topic: UNIX and Windows Operating Systems

Total MCQs: 30

☐ Answer key provided at the end

1. UNIX is primarily a(n):

A. Single-user OS

B. Multi-user, multitasking OS

- C. Real-time OS
- D. Single-tasking OS

☐ **Answer: B**

2. Windows was developed by:

- A. Apple
- B. IBM
- C. Microsoft
- D. Google

☐ **Answer: C**

3. What is the default shell in most UNIX systems?

- A. command.com
- B. bash
- C. explorer
- D. powershell

☐ **Answer: B**

4. In Windows, the main system file manager is:

- A. cmd.exe
- B. bash
- C. explorer.exe
- D. terminal.exe

☐ **Answer: C**

5. Which of the following is a valid UNIX command to list files?

- A. DIR
- B. ls
- C. list
- D. show

☐ **Answer: B**

6. In Windows, the command dir is used to:

- A. Change directory
- B. Delete file
- C. List files
- D. Show date

☐ **Answer: C**

7. Which operating system uses case-sensitive file names?

- A. Windows
- B. UNIX
- C. DOS
- D. Android

☐ **Answer: B**

8. Windows is based on which interface?

- A. CLI
- B. TUI
- C. GUI
- D. Text-based

☐ **Answer: C**

9. The command to shut down a UNIX system is:

- A. halt
- B. shutdown
- C. poweroff
- D. All of the above

☐ **Answer: D**

10. Which file system is commonly used in UNIX?

- A. FAT32
- B. NTFS
- C. ext4

D. HFS+

☐ Answer: C

11. What is the superuser in UNIX called?

A. root

B. admin

C. sudo

D. user0

☐ Answer: A

12. chmod command in UNIX is used to:

A. Change directory

B. Modify file contents

C. Change file permissions

D. Exit terminal

☐ Answer: C

13. In Windows, file extensions help to:

A. Identify the type of file

B. Run a program

C. Display a GUI

D. Manage memory

☐ Answer: A

14. Which is the default drive for Windows installations?

A. A:

B. B:

C. C:

D. D:

☐ Answer: C

15. The Windows Registry stores:

- A. Temporary files
- B. Hardware and software settings
- C. User documents
- D. Log files

☐ **Answer: B**

16. Which command is used in UNIX to view the content of a file?

- A. view
- B. ls
- C. type
- D. cat

☐ **Answer: D**

17. The Windows equivalent of UNIX mv command is:

- A. move
- B. copy
- C. rename
- D. paste

☐ **Answer: A**

18. Which Windows version introduced the NTFS file system?

- A. Windows 95
- B. Windows NT
- C. Windows XP
- D. Windows 98

☐ **Answer: B**

19. The grep command in UNIX is used for:

- A. Sorting files
- B. Searching text in files
- C. Deleting files

D. Renaming files

☐ Answer: B

20. What is the Windows equivalent of grep?

A. findstr

B. search

C. scan

D. grep32

☐ Answer: A

21. In UNIX, which command shows the current directory?

A. dir

B. where

C. pwd

D. ls

☐ Answer: C

22. Windows powershell is:

A. UNIX shell

B. Legacy shell

C. Task manager

D. Scripting and automation shell

☐ Answer: D

23. Windows is:

A. Open-source

B. Freeware

C. Closed-source and licensed

D. Only for servers

☐ Answer: C

24. Which of the following is NOT a valid Windows file extension?

- A. .exe
- B. .bat
- C. .sh
- D. .docx

☐ **Answer: C**

25. UNIX file permissions include:

- A. Read, Write, Execute
- B. Copy, Paste, Delete
- C. View, Rename, Remove
- D. Load, Save, Move

☐ **Answer: A**

26. In UNIX, kill command is used to:

- A. Stop a process
- B. Delete files
- C. Logout
- D. Reboot

☐ **Answer: A**

27. Windows Task Manager is used to:

- A. Create files
- B. Monitor system performance and processes
- C. Install software
- D. Format drives

☐ **Answer: B**

28. In UNIX, the symbol ~ represents:

- A. Current directory
- B. Root directory
- C. Home directory

D. Temporary files

☐ **Answer: C**

29. The ps command in UNIX is used to:

A. Print screen

B. Display processes

C. Start shell

D. Set permissions

☐ **Answer: B**

30. Windows OS supports which file systems?

A. NTFS

B. FAT32

C. exFAT

D. All of the above

☐ **Answer: D**
